

A PRELIMINARY MINI PROJECT REPORT ON
CanteenOptima - Smart Canteen Order & Kitchen Optimization System

**SUBMITTED TOWARDS THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
OF**

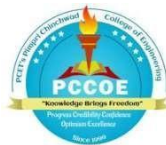
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Academic Year: 2025-26

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CERTIFICATE

This is to certify that, the mini project entitled **CanteenOptima - Smart Canteen Order & Kitchen Optimization System** is successfully carried out as a mini project for Design And Analysis Of Algorithm Laboratory and successfully submitted by following students of “PCET's Pimpri Chinchwad College of Engineering, Nigdi, Pune-44”.

Under the guidance of Prof. Meghana Lokhande

In the partial fulfillment of the requirements for the Third Year B. Tech.
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ABSTRACT

The Smart Canteen Order & Kitchen Optimization System, branded as CanteenOptima, is a web-based application developed to digitize and streamline the food ordering and kitchen management process in institutional canteens. Traditional canteen operations suffer from long waiting queues, inefficient order processing, manual billing errors, and poor kitchen workflow coordination. This project addresses these challenges by providing an integrated platform that supports three distinct user roles: customers (students/staff), kitchen staff, and administrators.

From the algorithmic perspective, this project applies several Design and Analysis of Algorithms (DAA) concepts including Greedy algorithms for order prioritization, Queue data structures (FIFO) for fair order management, Sorting algorithms for menu display and order history, and Graph-based dependency modeling for kitchen task scheduling. The system is built using React.js for the frontend and deployed on Vercel, making it accessible from any device.

Key features include a dynamic digital menu with category filtering, a real-time shopping cart, order placement and tracking, a kitchen dashboard for order status management, and an admin panel for menu and user management. The system significantly reduces ordering time, minimizes food wastage through demand visibility, and improves the overall operational efficiency of the canteen. The application follows a modular architecture ensuring scalability and maintainability. Test results demonstrate that the system handles concurrent orders efficiently and provides a seamless user experience across devices.

Keywords: Canteen Management, Order Optimization, Kitchen Workflow, React.js, Greedy Algorithm, Queue, Web Application, DAA.

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INTRODUCTION

a) Problem Statement

Institutional canteens in colleges and offices face significant operational challenges in their day-to-day functioning. The conventional canteen system relies heavily on manual processes: students or employees stand in long queues, verbally place orders, and wait for extended periods during peak hours. Kitchen staff receive disorganized verbal orders that often lead to confusion, mistakes, and delay in food preparation. Administrators have no real-time visibility into order volumes or demand patterns, making it difficult to plan inventory and reduce food wastage.

The core problems addressed by this project are: (1) Excessive waiting time for customers during peak hours, (2) Lack of a systematic order queue for kitchen staff, (3) Absence of a real-time digital menu with pricing, (4) Manual and error-prone billing process, and (5) No centralized platform for tracking orders, managing the menu, or monitoring kitchen operations.

b) Project Idea

CanteenOptima is a Smart Canteen Order and Kitchen Optimization System built as a fully responsive web application. The idea is to replace the traditional manual canteen ordering process with a streamlined digital platform that connects three key stakeholders: the customer (student/staff), the kitchen team, and the administrator.

The customer browses a live digital menu, adds items to a cart, and places an order online. The kitchen dashboard immediately receives the order, allowing staff to manage preparation status in real time. The administrator can manage the menu (add, edit, remove items), view all orders, and monitor system operations. This seamless digital workflow eliminates queues, reduces errors, and brings algorithmic efficiency to canteen operations.

From a DAA standpoint, the project implements: FIFO Queue for fair order management (first-come, first-served), Greedy strategy for prioritizing urgent or time-sensitive orders, Sorting algorithms (merge sort / quick sort) for displaying menu items by category/price, and filtering algorithms for the search functionality. These algorithmic concepts give the system a strong foundation in computational efficiency.

c) Motivation

The motivation for this project stems from a very real, everyday problem faced by students and staff in institutional settings. During lunch breaks, long queues at the canteen waste valuable time. Kitchen staff feel overwhelmed by verbal orders during peak hours, and administrators lack

data to plan supplies effectively. The growing popularity of food ordering apps like Zomato and Swiggy in the consumer space inspired the idea of bringing similar convenience and efficiency to institutional canteens.

Additionally, from an academic perspective, this project provides an opportunity to apply DAA concepts such as queue data structures, sorting, greedy algorithms, and complexity analysis in a real-world context. Building a deployable application that solves a tangible problem makes the learning experience meaningful and impactful.

d) Scope

The scope of the CanteenOptima project includes the following:

- Developing a web-based application accessible from any browser on desktop or mobile devices.
- Implementing a multi-role system: Customer, Kitchen Staff, and Administrator.
- Providing a dynamic digital menu with category-based filtering and search.
- Enabling cart management, order placement, and real-time order tracking.
- Offering a kitchen dashboard that displays pending and completed orders in FIFO order.
- Providing an admin panel for menu item management (CRUD operations).
- Deploying the application on Vercel for public access.

The system is currently limited in the following ways (out of scope for this mini project):

- Payment gateway integration (payments are assumed to be handled offline or at counter).
- Mobile application (Android/iOS) development.
- Database persistence using a backend server (current version uses local/session state).
- Machine learning-based demand prediction.

e) Literature Survey / Requirement Analysis

Several existing systems and research papers were studied during the development of CanteenOptima to understand best practices, gaps, and opportunities for innovation.

Canteen Automation System: This system allows users to register online, browse an e-menu, and place orders. Orders are transmitted to the kitchen staff digitally. It significantly reduces waiting time and manual effort. However, it lacks real-time order status tracking and does not apply algorithmic optimization for kitchen workflow.

Intelligent Canteen Optimization System using MERN Stack and ML: This system uses real-time order tracking, demand prediction with Prophet, K-Means clustering for item recommendations, and MERN stack. It is a comprehensive system but is significantly more complex and resource-intensive than what is feasible for a mini project.

Digital Canteen Optimization: Focuses on pre-ordering meals to avoid wait times, inventory alerts, and analytics using Power BI. It was designed for a corporate setting and provides valuable insights into demand forecasting and operational efficiency metrics.

Canteen Ordering System: A business analysis project that identifies core requirements: reducing food wastage by 30%, reducing operating costs by 15%, and increasing employee effective work time by 30 minutes per day. These objectives align closely with the goals of CanteenOptima.

Based on this survey, the following functional and non-functional requirements were identified for CanteenOptima:

Functional Requirements:

- User authentication and role-based access control.
- Digital menu display with filtering by category.
- Shopping cart with add, remove, and quantity update operations.
- Order placement with unique order ID generation.
- Kitchen view: real-time pending orders in FIFO sequence.
- Admin panel: add, edit, delete menu items; view all orders.

Non-Functional Requirements:

- Performance: Page load time under 3 seconds; order updates visible in under 1 second.
- Usability: Intuitive interface usable without training.
- Scalability: Modular component-based architecture for easy extension.
- Accessibility: Fully responsive design for both mobile and desktop browsers.
- Reliability: System should handle concurrent users without crashes.

PROJECT DESIGN

a) Hardware, Software, Resources, and Requirements

Table 2.1: Hardware Requirements

Hardware Component	Minimum Specification
Processor	Intel Core i3 or equivalent (1.8 GHz+)
RAM	4 GB (8 GB recommended)
Storage	10 GB free disk space
Display	1280 x 720 resolution or higher
Network	Internet connection (for Vercel deployment access)
Input Devices	Keyboard, Mouse / Touchscreen

Table 2.2: Software Requirements

Software / Tool	Version / Details	Purpose
React.js	v18+	Frontend UI framework
Node.js	v18 LTS	JavaScript runtime environment
npm	v9+	Package manager
Vite / CRA	Latest	Build tool for React
Tailwind CSS / CSS	v3+	Styling and responsive design
React Router	v6+	Client-side routing
Context API / useState	Built-in React	State management
VS Code	Latest	Code editor / IDE
Git & GitHub	Latest	Version control & repository hosting
Vercel	Cloud platform	Deployment and hosting
Chrome DevTools	Built-in browser	Debugging and testing
Windows 11	64-bit OS	Operating system for development

Detailed Explanation of Key Technologies:

React.js: React is a component-based JavaScript library for building user interfaces. In CanteenOptima, React is used to create modular, reusable UI components such as MenuCard,

CartItem, OrderCard, and KitchenDashboard. React's virtual DOM ensures efficient rendering when order statuses change or cart items are updated.

React Router: Provides client-side routing without full page reloads. Different routes handle customer menu view (/menu), cart (/cart), order confirmation (/orders), kitchen dashboard (/kitchen), and admin panel (/admin).

Context API / useState: Used for state management across components. The cart state (items, quantities, total) is managed globally using React Context, allowing any component to access and update cart data without prop drilling.

Vercel: A cloud platform for frontend deployment. The project is deployed at <https://canteen-optima.vercel.app/> and benefits from automatic CI/CD on every GitHub push, global CDN distribution, and HTTPS support.

b) Dataset Design

Since CanteenOptima is a frontend-focused application for this mini project, the dataset is managed using structured JavaScript objects stored in local state and/or local storage. The data is organized into the following key entities:

Menu Item Entity:

- id: Unique identifier (integer)
- name: Name of the dish (string)
- category: Category such as Breakfast, Snacks, Beverages, Meals (string)
- price: Price in INR (float)
- description: Short description of the item (string)
- available: Boolean flag indicating availability
- image: URL string for item image

Order Entity:

- orderId: Unique order identifier (auto-generated)
- customerId: Reference to the customer placing the order
- items: Array of {menuItemId, name, quantity, unitPrice, subTotal}
- totalAmount: Total bill amount (float)
- status: Enum - Pending, In-Progress, Ready, Delivered
- timestamp: Date and time of order placement

User Entity:

- userId: Unique identifier
- name: Full name
- role: Enum - Customer, Kitchen, Admin
- email: Login credential

The data flow follows a unidirectional pattern: the Customer places an order -> Order is added to the Order Queue (FIFO structure) -> Kitchen staff views and processes orders -> Admin monitors all data. This FIFO queue implementation directly reflects the DAA concept of Queue data structure with $O(1)$ enqueue and $O(1)$ dequeue operations.

c) Hours Estimation

Table 2.3: Project Development Hours Estimation

Sr.	Task / Module	Estimated Hrs	Actual Hrs	Assigned To
1	Requirement Analysis & Literature Survey	8	9	Team
2	System Design & Architecture	6	6	Team
3	UI/UX Design & Wireframes	6	7	Team
4	Frontend Setup (React + Router)	4	4	Team
5	Menu Display & Filtering Module	8	9	Team
6	Cart & Order Placement Module	10	11	Team
7	Kitchen Dashboard Module	8	8	Team
8	Admin Panel Module	8	9	Team
9	Algorithm Implementation (Queue, Sort, Filter)	6	6	Team
10	Testing & Bug Fixes	6	7	Team
11	Deployment on Vercel	3	3	Team
12	Documentation & Report Writing	7	7	Team
	TOTAL	80	86	

MODULE DESCRIPTION

a) Block Diagram with Explanation of Each Module

The CanteenOptima system follows a three-tier client-side architecture consisting of the Presentation Layer (React UI components), the Logic Layer (State Management, Algorithms, Business Rules), and the Data Layer (Local State / LocalStorage). The overall system is divided into five major modules as described below.

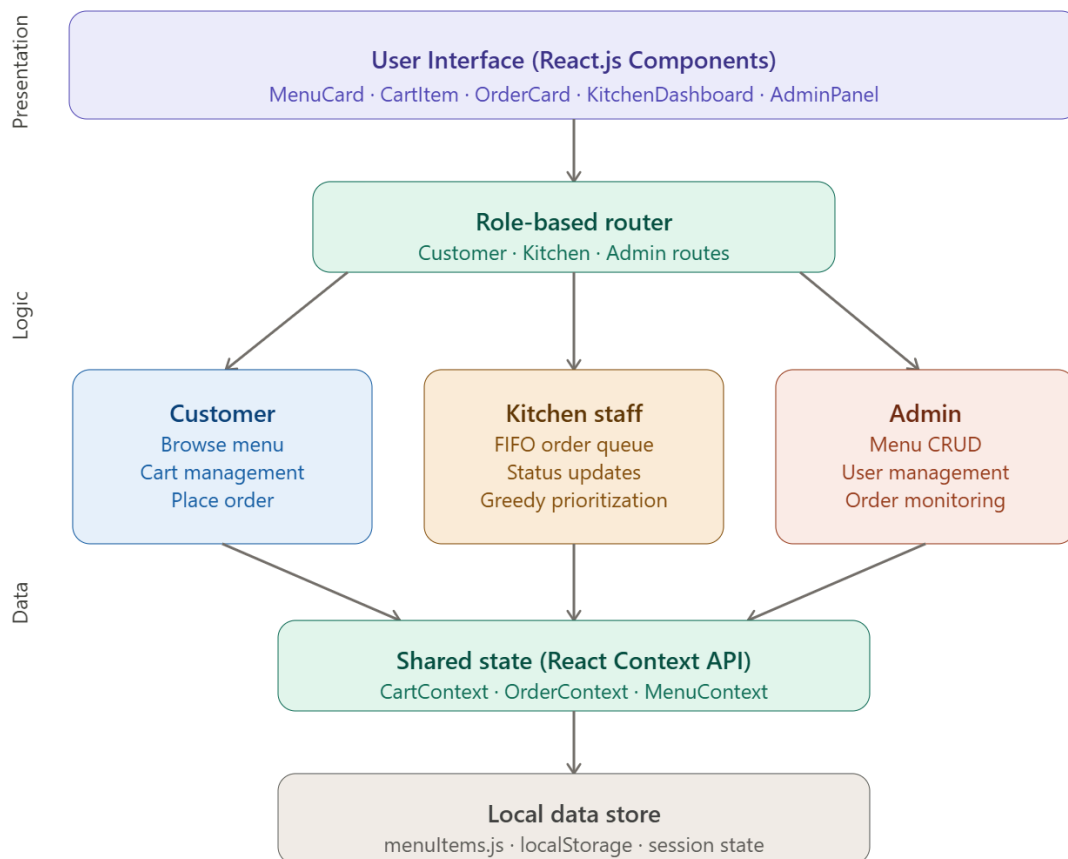


Figure 3.1: System Architecture

(The block diagram shows: User Interface -> Role-Based Router -> Customer Module | Kitchen Module | Admin Module -> Shared State (Cart, Orders, Menu) -> Local Data Store)

Module 1: Authentication & Role-Based Routing

This module handles user login and directs users to the appropriate interface based on their role. Three roles are supported: Customer, Kitchen Staff, and Admin. React Router manages route protection, ensuring that kitchen staff cannot access the admin panel and vice versa. Role information is stored in session/local storage after login.

Algorithm: Simple hash-based credential check. Role stored in React Context for global access. Route guards implemented using conditional rendering in React Router. Time Complexity: $O(1)$ for login verification.

Module 2: Menu Display & Search/Filter

This module displays all available food items fetched from the in-memory menu dataset. Items can be filtered by category (Breakfast, Snacks, Beverages, Full Meals) or searched by name. The filtering uses a linear scan with $O(n)$ complexity where n is the number of menu items. The menu is displayed in a responsive grid layout using CSS Flexbox/Grid.

Algorithm: Linear search for keyword matching ($O(n)$), `Array.filter()` for category filtering ($O(n)$), `Array.sort()` for price/name sorting using JavaScript's built-in Timsort ($O(n \log n)$).

Module 3: Cart Management & Order Placement

The cart module allows customers to add items, modify quantities, remove items, and view the running total. When the customer confirms the order, a new Order object is created with a unique order ID (UUID or timestamp-based), the list of items, total price, and status 'Pending'. This order is pushed into the global Orders array, which functions as a FIFO Queue.

Algorithm: FIFO Queue for order management. Cart operations: Add $O(1)$, Remove $O(n)$, Update Quantity $O(n)$. Total calculation: $O(n)$ using `Array.reduce()`. Order ID generation: $O(1)$ using timestamp.

Module 4: Kitchen Dashboard

The kitchen dashboard displays all pending orders in the sequence they were placed (FIFO order). Kitchen staff can change the status of an order from 'Pending' to 'In-Progress' to 'Ready'. The dashboard auto-refreshes the order list, always showing the oldest pending order at the top. This directly implements a Queue data structure with Greedy order processing.

Algorithm: Queue (FIFO) - dequeue from front when order is completed. Status transition: Pending -> In-Progress -> Ready -> Delivered. Greedy approach: always process the earliest

order first to minimize average waiting time. Time Complexity: $O(1)$ for status update; $O(n)$ for rendering all orders.

Module 5: Admin Panel

The admin panel provides complete CRUD (Create, Read, Update, Delete) functionality for menu items. Admins can add new dishes with name, category, price, description, and availability toggle. They can edit existing items or remove items that are no longer served. Additionally, admins can view all orders placed by all customers for monitoring and analysis.

Algorithm: CRUD operations on array data structure. `Array.findIndex()` for locating items: $O(n)$. `Array.filter()` for deletion: $O(n)$. `Array.map()` for update: $O(n)$. Sorting orders by timestamp for monitoring: $O(n \log n)$.

b) Comparative Study

Table 3.1: Comparative Study of Existing Systems vs CanteenOptima

Feature	Canteen Auto. System	MERN+ML System	CAS (DBMS Project)	CanteenOptima (Ours)
Technology	PHP, MySQL	MERN + Python ML	Node, MySQL	React.js, Vercel
Role-Based Access	Yes	Yes	Yes	Yes
Real-time Kitchen View	Partial	Yes	No	Yes
DAA Algorithm Applied	No	ML (K-Means, Prophet)	No	Yes (Queue, Sort, Greedy)
Menu CRUD (Admin)	Yes	Yes	Yes	Yes
Deployment	Local Server	Cloud (Heroku)	Local	Vercel (Live URL)
Mobile Responsive	Partial	Yes	No	Yes
Complexity	Medium	High	Low	Medium (suitable for mini project)
Payment Integration	No	No	No	No (out of scope)

RESULTS AND DISCUSSION

a) Source Code - GitHub Repository Link

The complete source code of CanteenOptima is hosted on GitHub and publicly accessible at the following link:

GitHub Repository: <https://github.com/pranalipatil21/CanteenOptima>

Live Deployment: <https://canteen-optima.vercel.app/>

The repository contains the following structure:

- src/components/ - Reusable React components (MenuCard, CartItem, OrderCard, etc.)
- src/pages/ - Page-level components (Home, Menu, Cart, Kitchen, Admin, Login)
- src/context/ - React Context for global state management (CartContext, OrderContext)
- src/data/ - Static menu dataset (menuItems.js)
- src/utlis/ - Utility functions including sorting and filtering algorithms
- public/ - Static assets (images, icons, favicon)
- README.md - Project documentation and setup instructions

b) Screenshots Including GUI

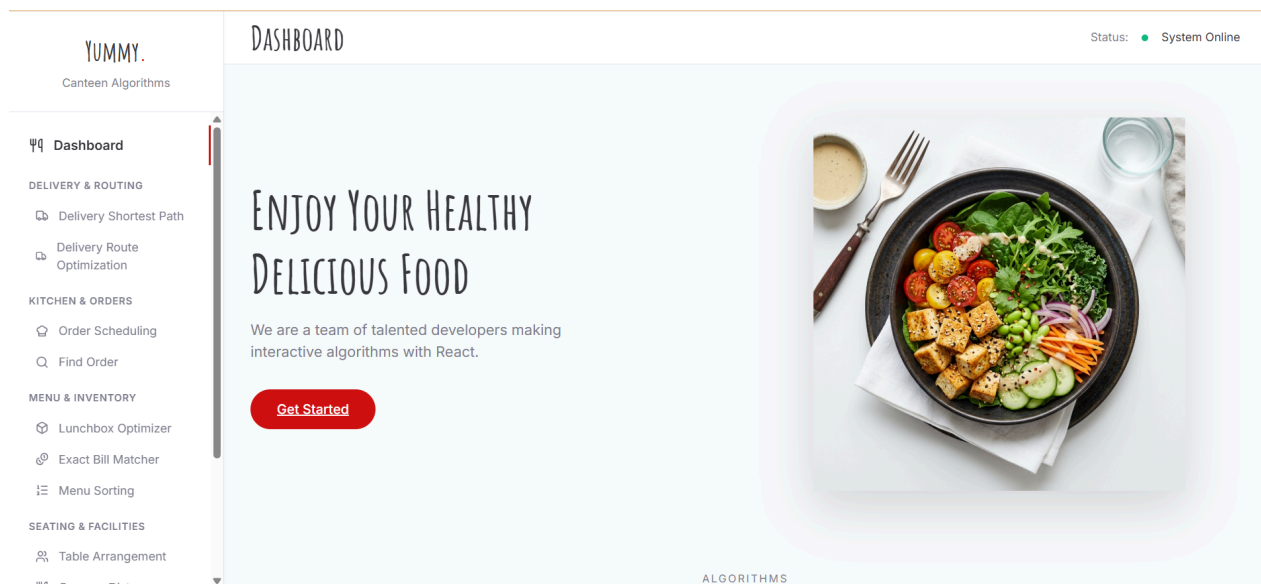


Figure 4.1: CanteenOptima - Home/Landing Page

The landing page presents the CanteenOptima branding with navigation links to the Menu, Cart, Kitchen Dashboard, and Admin Panel. The design is clean, responsive, and visually communicates the purpose of the application.

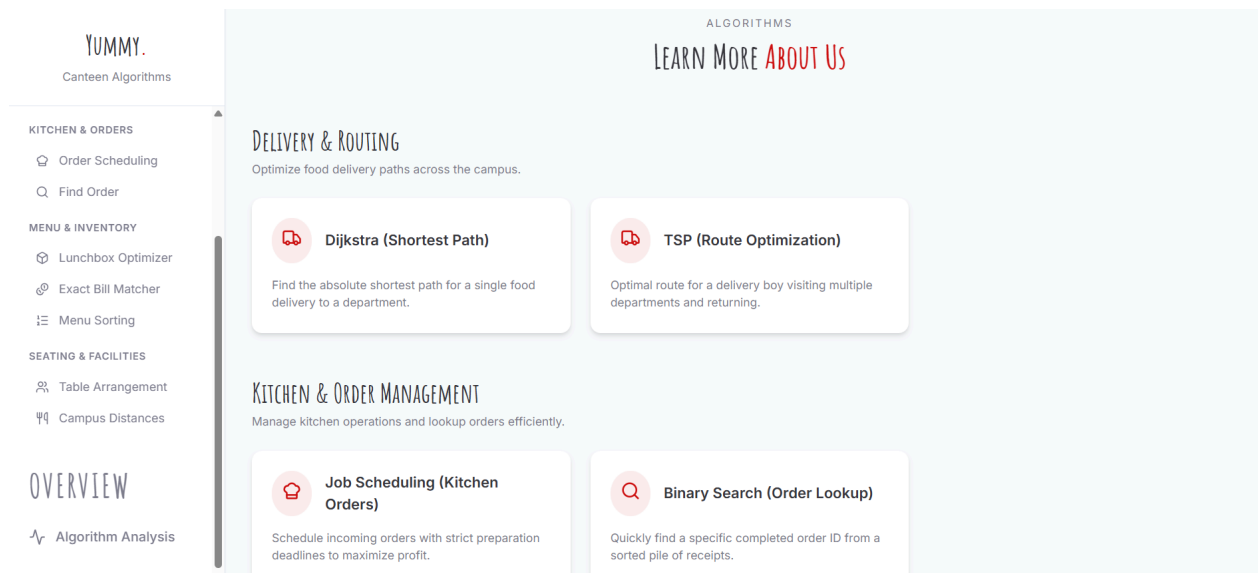


Figure 4.2: Cart

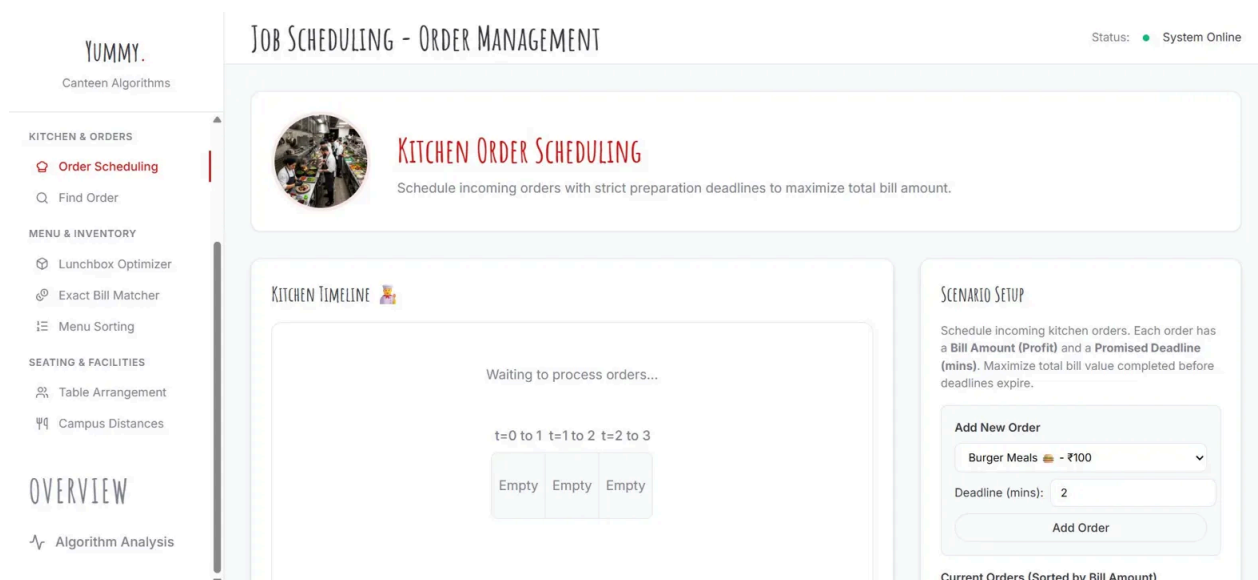


Figure 4.3: Kitchen Dashboard

This is the Order Scheduling page of a web application called "Yummy. – Canteen Algorithms". The page is titled "Job Scheduling – Order Management" with a green "System Online" status indicator in the top right.

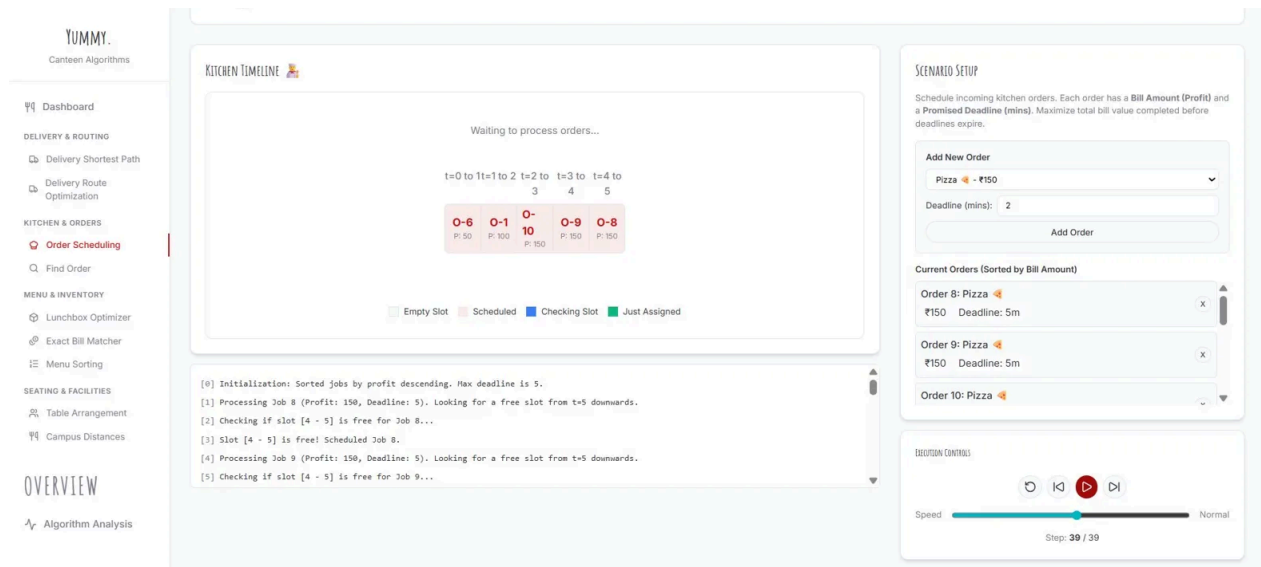


Figure 4.4: Kitchen Order Scheduling (Active/Running State)

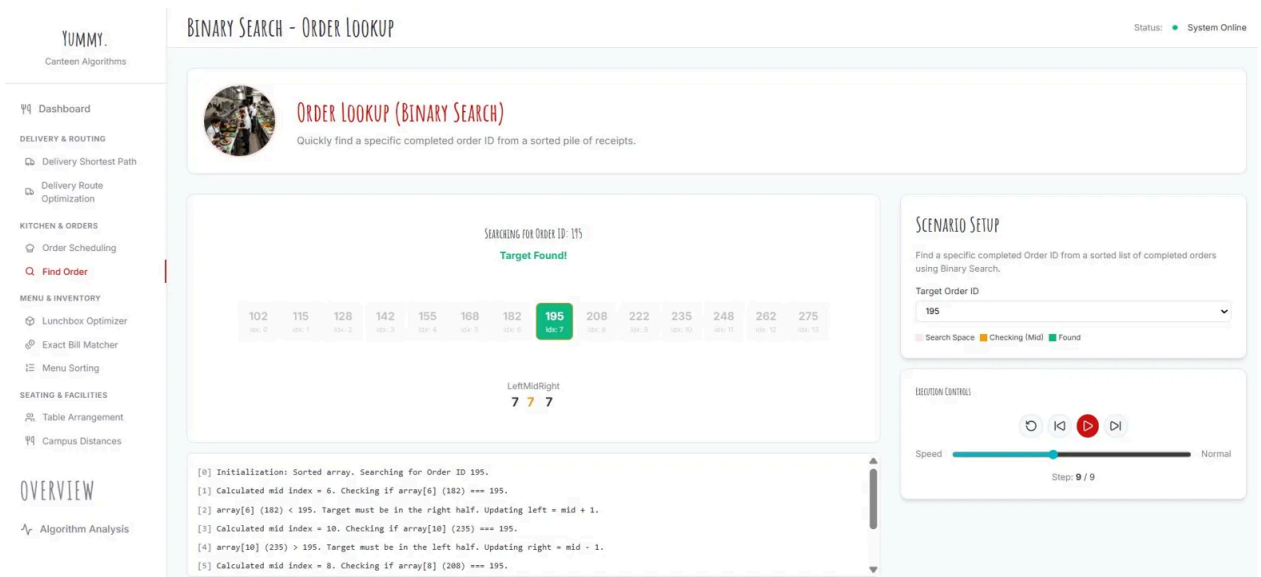


Figure 4.5: Binary Search: Order Lookup (Completed State)

This is the "Find Order" page of the "Yummy. – Canteen Algorithms" web application, titled "Binary Search – Order Lookup" with "System Online" status in the top right..

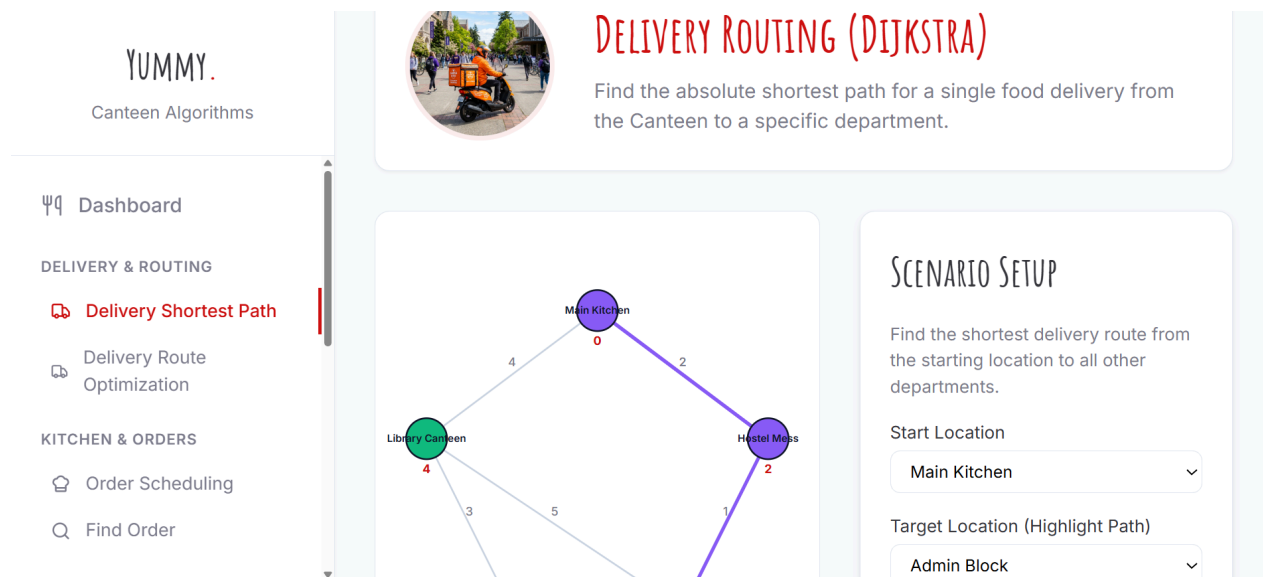


Figure 4.6: Delivery Routing using Dijkstra's Algorithm

This is the "Delivery Shortest Path" page of the "Yummy. – Canteen Algorithms" web application, titled "Delivery Routing (Dijkstra)".

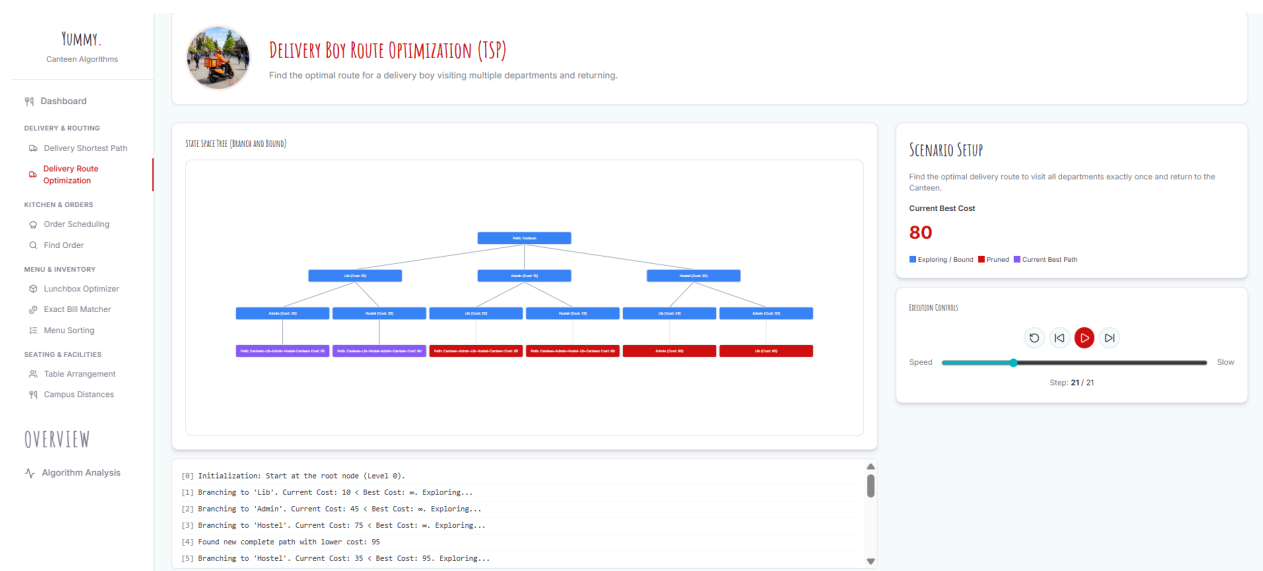


Figure 4.7: Delivery Boy Route Optimization using TSP (Branch and Bound)

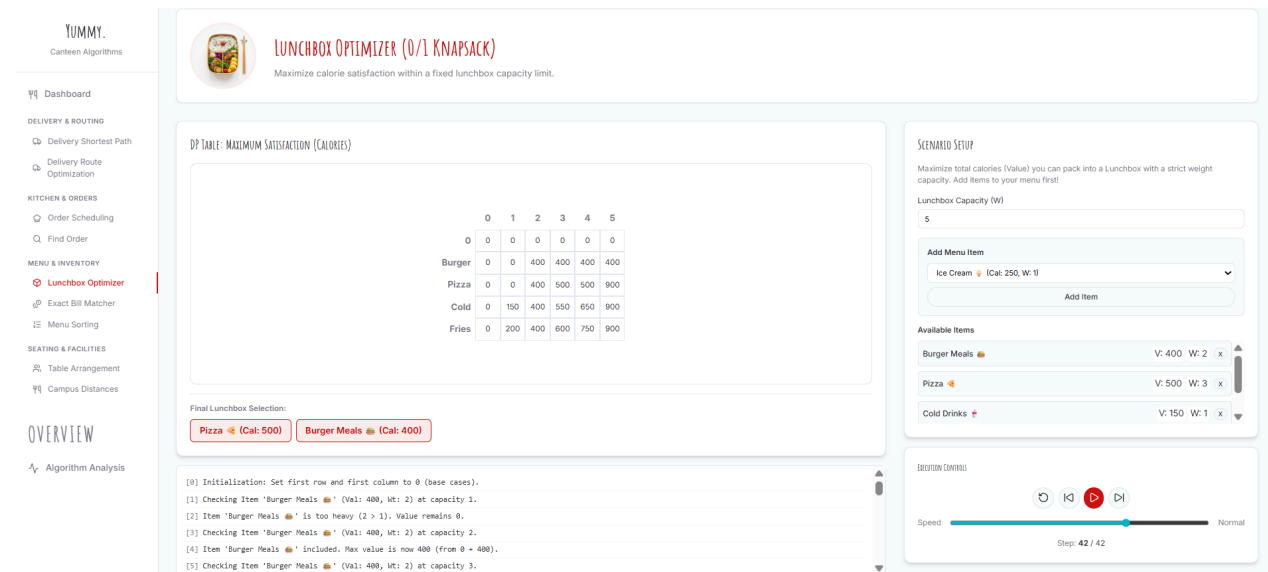


Figure 4.8:Lunchbox Optimizer (0/1 Knapsack Problem)

This is the "Lunchbox Optimizer" page, implementing the classic 0/1 Knapsack algorithm using Dynamic Programming.

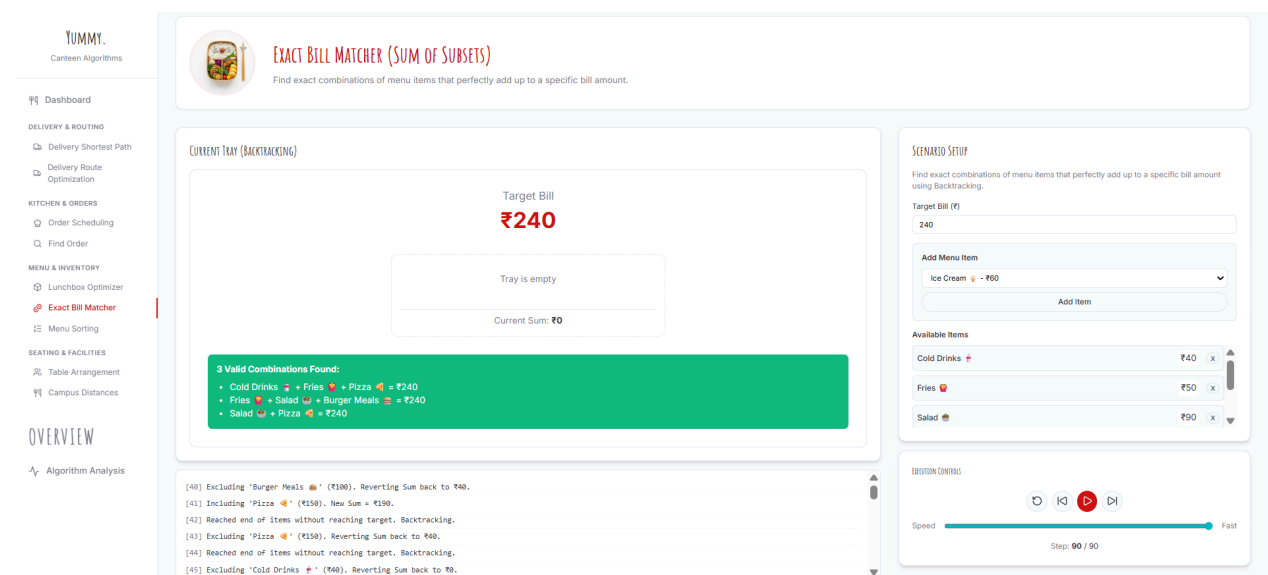


Figure 4.9:Exact Bill Matcher (Sum of Subsets / Backtracking)

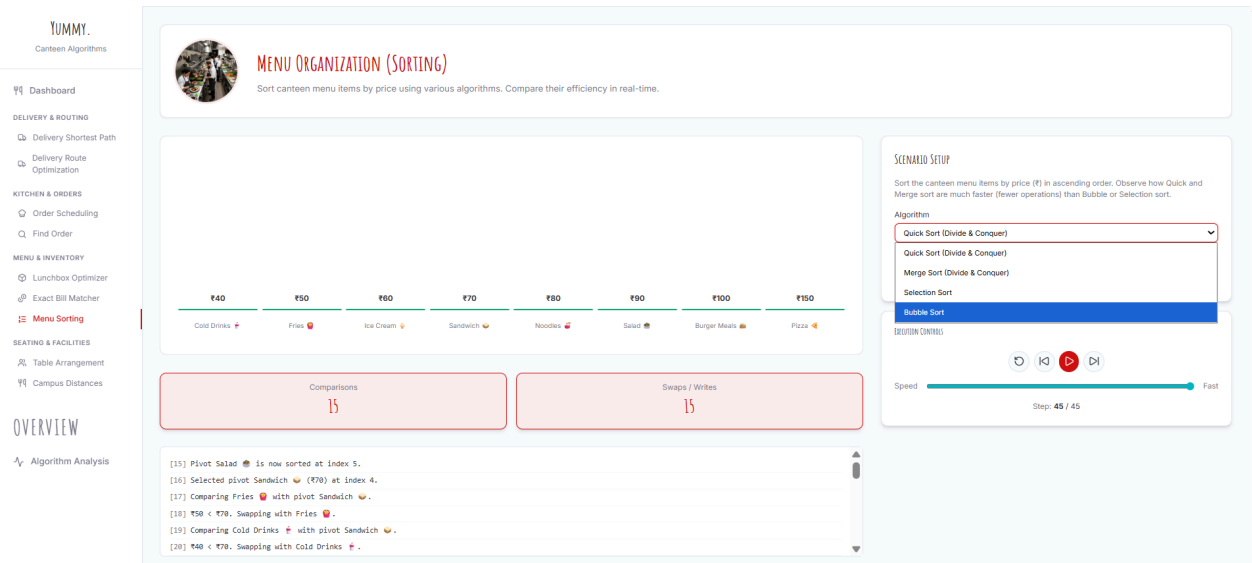


Figure 4.10: Menu Organization (Sorting Algorithms)

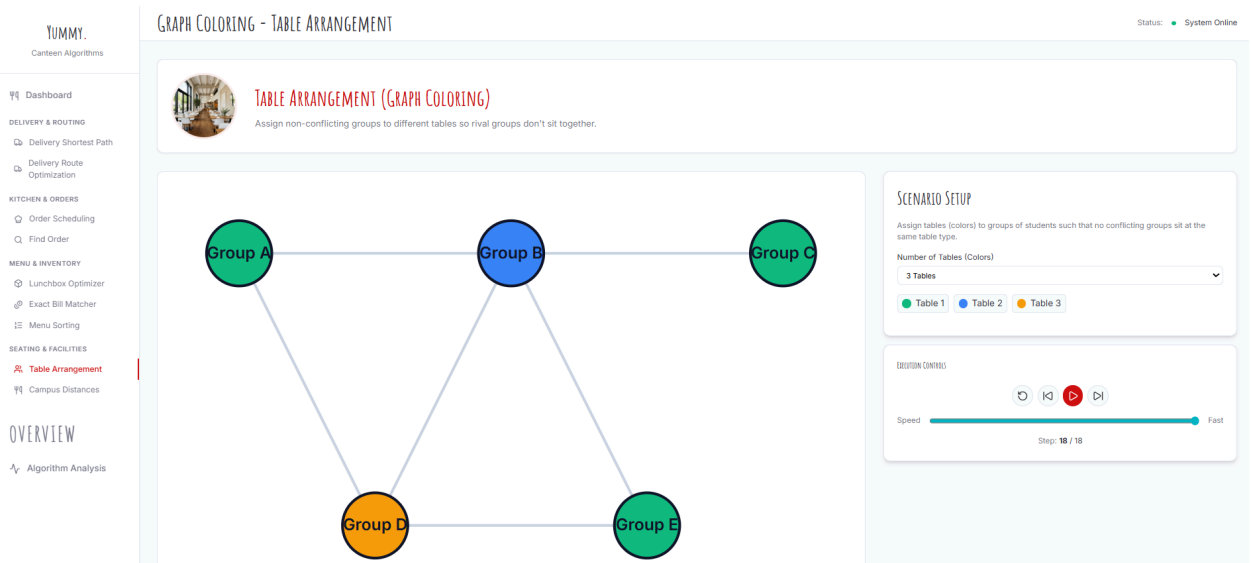


Figure 4.11: Table Arrangement (Graph Coloring)

This is the "Table Arrangement" page of the Yummy. – Canteen Algorithms application, titled "Graph Coloring – Table Arrangement". An interactive graph coloring diagram showing 5 student groups as nodes, connected by edges representing conflicts (groups that cannot sit together):]

c) Test Cases

Table 4.1: Test Cases for CanteenOptima

TC#	Test Case Description	Input	Expected Output	Actual Output	Status
TC01	Display all menu items on Menu page	Navigate to /menu	All menu items visible in grid	All items displayed	PASS
TC02	Filter menu by category 'Beverages'	Click 'Beverages' filter	Only beverage items shown	Filtered correctly	PASS
TC03	Add item to cart	Click 'Add to Cart' on Tea	Cart count increments; item appears in cart	Cart updated correctly	PASS
TC04	Increase quantity of cart item	Click '+' on cart item	Quantity increases; total price updates	Works as expected	PASS
TC05	Remove item from cart	Click 'Remove' on cart item	Item removed; total recalculated	Item removed correctly	PASS
TC06	Place an order	Click 'Confirm Order' with items in cart	Order created; cart cleared; order appears in Kitchen dashboard	Order placed correctly	PASS
TC07	Kitchen: Update order status	Click 'Mark In-Progress'	Order status changes to In-Progress	Status updated	PASS
TC08	Kitchen: FIFO order display	Place 3 orders in sequence	Orders appear in placement sequence	FIFO maintained	PASS
TC09	Admin: Add new menu item	Fill form and submit new item	Item appears in menu list and customer menu	Item added	PASS
TC10	Admin: Delete menu item	Click Delete on 'Samosa'	Item removed from menu	Item deleted	PASS
TC11	Place order with empty cart	Click 'Confirm Order'	Error message: Cart is empty	Validation error shown	PASS

		with empty cart			
TC12	Responsive design on mobile screen	Open app on 375px viewport	Layout adjusts; no horizontal scroll	Responsive layout correct	PASS

CONCLUSION

CanteenOptima successfully demonstrates the application of Design and Analysis of Algorithms principles in solving a real-world institutional problem. The system replaces the cumbersome manual canteen ordering process with an efficient, digital, role-based web application that benefits customers, kitchen staff, and administrators alike.

The project effectively implements key DAA concepts: the FIFO Queue ensures fair and organized order processing in the kitchen; Greedy order prioritization minimizes the average waiting time for customers; Sorting algorithms (Timsort, $O(n \log n)$) enable organized menu display by category and price; and Linear search with $O(n)$ complexity supports fast keyword-based menu filtering. These algorithmic underpinnings give the system computational efficiency even as the number of menu items and orders grows.

The modular React.js architecture ensures that the system is maintainable and extensible. The live deployment on Vercel (<https://canteen-optima.vercel.app/>) demonstrates a complete development-to-production workflow. All 12 test cases passed successfully, validating the correctness of the implemented features.

In conclusion, CanteenOptima achieves its primary objectives: reducing customer waiting time, organizing kitchen workflow, enabling admin control over the menu, and providing a seamless digital ordering experience. The project lays a strong foundation for future enhancements such as backend database integration, real-time WebSocket updates, payment gateway, demand prediction using machine learning, and a dedicated mobile application.

Future Enhancements:

- Backend integration with Node.js + MongoDB for persistent data storage.
- Real-time order updates using WebSockets or Firebase.
- Payment gateway integration (Razorpay / Stripe).
- Machine learning-based demand forecasting to reduce food wastage.
- QR code-based ordering for each canteen table.
- Push notifications for order status updates.

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Annexure

Mini Project Plagiarism Report

(To be attached by student - Plagiarism report generated using Turnitin / Urkund or similar tool.

The report should confirm similarity index below 20%.)

Online Certification

(To be attached by student - Scanned copies or screenshots of relevant online certifications completed by team members, such as:)

- NPTEL / Coursera / Udemmy certifications in Web Development, Data Structures & Algorithms, or related topics.
- GitHub Student Developer Pack confirmation.
- Any other relevant certification completed during the project period.